

The first canonical window, its full phase product, and both arithmetic block budgets

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This chapter joins the original first-degree source calculation (CV.11)–(CV.12) and (TVB.37)–(TVB.38) to the consecutive-window proof of PR 25 at the exact head 66fac5e7885a40cd4873e74b754907320f05b067. The complete original and proposed corrected sources accompany this chapter. Their correction retains all zero generalized eigenvalues. The source radius, original norm envelope and finite propagation are proved below with their original domains. The full first-window phase product has a single terminal contraction and two copies of each interior contraction; its first step contains no preceding quotient inverse.

1 Original source and the actual first-degree step

Retain $g = 2\xi$, the complete fixed packet h , integer tensor degree $k \geq 1$, and the original density

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad S = c + iu, \quad c = k/2. \quad (\text{CJ.1})$$

Its source inner product is the integral against $m_{h,k}(u)du$; the original monic orthogonal polynomials p_n have squared norms $\omega_n > 0$, including $\omega_0 = \mu_h^k$. Use the original dagger-stable monic cyclic polynomial χ of degree $q \geq 1$, the full quotient $E = \mathbb{C}[S]/(\chi)$ and the original remainder basis. Its columns, kernels and metrics are

$$\begin{aligned} b_n &= [p_n]_\chi, \quad B_N = [b_0, \dots, b_N], \quad O_N = \text{diag}(\omega_0, \dots, \omega_N), \\ K_N &= B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N \quad (N \geq q-1). \end{aligned} \quad (\text{CJ.2})$$

Surjectivity of B_N and positivity of O_N prove $K_N > 0$. The section $R_N = O_N^{-1} B_N^* G_N$ satisfies $B_N R_N = I$ and $R_N^* O_N R_N = G_N$; its orthogonality to $\ker B_N$ proves that it is the unique minimum lift. Source inclusion proves $G_{N+1} \preceq G_N$ and $0 < V_{N+1} \leq V_N$.

Write the original expansion, retaining every coefficient,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle}{\omega_j}, \quad \nu_0 = \|\chi\|^2 = \omega_q + \sum_{j < q} |\gamma_j|^2 \omega_j. \quad (\text{CJ.3})$$

The first q remainder columns form an invertible triangular matrix, so $b_a^* G_{q-1} b_b = \delta_{ab} \omega_b$ for $a, b < q$. Reduction of (CJ.3) gives $b_q = -\sum_{j < q} \gamma_j b_j$ and hence $b_q^* G_{q-1} b_q = \nu_0 - \omega_q$. The exact rank-one determinant formula therefore gives

$$\delta_q := \frac{V_q}{V_{q-1}} = \left(1 + \frac{b_q^* G_{q-1} b_q}{\omega_q} \right)^{-1} = \frac{\omega_q}{\nu_0}. \quad (\text{CJ.4})$$

The rank-one determinant identity follows by taking a basis whose first vector is the update vector u when it is nonzero. In that basis uv^* has entries only in its first row; $I + uv^*$ is therefore upper triangular with diagonal entries $1 + v^*u, 1, \dots, 1$. Its determinant is $1 + v^*u$. The zero update gives the same identity directly.

Let ϕ_N denote the retained real phase of the original theta deformation, with the sign convention of (TVB.26): $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. Its reality follows in the literal chart $S = c + iu$ as follows. Write $Q_n(u) = i^{-n} p_n(c + iu)$ and $\hat{\chi}(u) = i^{-q} \chi(c + iu)$. Dagger stability gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$, so $\hat{\chi}$ is real monic. The source measure is real and positive; the monic orthogonality equations therefore give real Q_n . If E is the exact coefficient map $P(S) \mapsto P(c + iu)$, then $E b_n = i^n [Q_n]_{\hat{\chi}}$, while the transported kernel and metric are $E K_N E^*$ and $E^{-*} G_N E^{-1}$. The latter has real entries because the phases in each kernel summand occur together with their conjugates. Thus the original cross pairing is i times a real number, with its original factor ω_N retained. This proves the stated reality and specifies the exact chart map rather than changing the metric. At the first degree that identity reads $\gamma_{q-1} = -i\phi_{q-1}$. The original two-column Hermitian control formula (CV.4), before specialization, is

$$\epsilon_N^2 = \frac{a_N d_N - |z_N|^2}{\omega_N^2}, \quad a_N = b_N^* G_N b_N, \quad z_N = b_N^* G_N b_{N+1}, \quad d_N = b_{N+1}^* G_N b_{N+1}. \quad (\text{CJ.5})$$

Here z_N is purely imaginary in the stated dagger-stable source. The original boundary-control identity on (E, G_N) is

$$H_N = G_N^{-1} (A^* G_N + G_N A - k G_N) = \frac{b_{N+1} b_N^* G_N + b_N b_{N+1}^* G_N}{\omega_N}, \quad A = M_S.$$

For completeness this identity follows directly from the original monic recurrence. Self-adjointness of multiplication by u in the integral pairing and orthogonality give $u Q_n = Q_{n+1} + \beta_n Q_n + \alpha_n Q_{n-1}$, where $\beta_n \in \mathbb{R}$ and $\alpha_n = \omega_n / \omega_{n-1}$ for $n \geq 1$; the lower term is zero at $n = 0$. Indeed all coefficients below degree $n-1$ vanish by pairing with $u Q_j$ of degree at most $n-1$, and pairing with Q_{n-1} gives the displayed norm ratio. In the original S coordinates this is $S p_n = p_{n+1} + (c + i\beta_n) p_n - \alpha_n p_{n-1}$. Reduce it modulo χ and insert it into $A K_N + K_N A^* - k K_N$. The diagonal coefficients are $c + i\beta_n + c - i\beta_n - k = 0$. For each internal adjacent pair the coefficient is $\omega_n^{-1} - \alpha_{n+1} \omega_{n+1}^{-1} = 0$. The remaining two terms are $(b_{N+1} b_N^* + b_N b_{N+1}^*) / \omega_N$. Multiplication on the right by the original G_N proves the written boundary-control formula with its signs. It has rank at most two, is self-adjoint in G_N , and has trace $(z_N + \bar{z}_N) / \omega_N = 0$. Squaring its two summands and taking the trace gives $[z_N^2 + \bar{z}_N^2 + 2a_N d_N] / \omega_N^2 = 2(a_N d_N - |z_N|^2) / \omega_N^2$. For $q \geq 2$ its full spectrum is $\epsilon_N, -\epsilon_N$ and $q-2$ zeros, with $\epsilon_N \geq 0$; these first two entries may themselves be zero. For $q = 1$ trace zero makes the entire control zero. This proves (CJ.5), including the cross-pairing. At $N = q-1$ the three entries are respectively $\omega_{q-1}, -\gamma_{q-1} \omega_{q-1}, \nu_0 - \omega_q$. Substitution proves

$$\epsilon_{q-1}^2 + \phi_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}}. \quad (\text{CJ.6})$$

For $q = 1$, the lower projection in (CV.12) is the zero source and $\epsilon_{q-1} = 0$; the retained phase identity and (CJ.6) still hold. No G_{q-2} occurs in this calculation.

Define $\Lambda_N = \omega_N / V_N$. Combining (CJ.4)–(CJ.6) gives the exact first step

$$(\epsilon_{q-1}^2 + \phi_{q-1}^2) \Lambda_{q-1} = \Lambda_q (1 - \delta_q). \quad (\text{CJ.7})$$

Indeed its left side is $(\nu_0 - \omega_q) / V_{q-1}$, and $\nu_0 / V_{q-1} = \omega_q / V_q$ by (CJ.4). This proves the identity with its single contraction and full original phase.

2 The complete first-window product and both finite budgets

At each regular degree $N \geq q$, the proved original radius identity (EP.11), equivalently (TVB.27), is

$$(\epsilon_N^2 + \phi_N^2) \Lambda_N = \Lambda_{N+1} (1 - \delta_N) (1 - \delta_{N+1}), \quad \delta_N = V_N / V_{N-1}. \quad (\text{CJ.8})$$

Every δ_N lies in $(0, 1]$. Multiply (CJ.7) by the consecutive identities (CJ.8) at $N = q, \dots, j-1$. All Λ_N are positive, so canceling their identical intermediate factors is valid. It yields, for every $j \geq q$,

$$\prod_{N=q-1}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_{q-1}}{\omega_{q-1} V_j} (1 - \delta_j) \prod_{N=q}^{j-1} (1 - \delta_N)^2. \quad (\text{CJ.9})$$

For $j = q$ the interior product is empty and equal to one, so this is precisely the first step. For $j > q$, each δ_N , $q \leq N < j$, occurs once from the step ending at N and once from the step beginning there. The terminal δ_j occurs once. This proves every exponent in (CJ.9). For a regular initial degree $i \geq q$, the identical multiplication instead gives

$$\prod_{N=i}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_i}{\omega_i V_j} (1 - \delta_i)(1 - \delta_j) \prod_{N=i+1}^{j-1} (1 - \delta_N)^2, \quad j > i. \quad (\text{CJ.10})$$

These two expressions explicitly relate the first and regular windows to the same source, phase and quotient volumes.

For the original off-line quartet let its common multiplicity be $m \geq 1$, its real displacement be $\delta > 0$ and its positive imaginary coordinate be $\gamma > 0$. The complete tensor calculation gives, with no omitted repeated root,

$$q = [1 + k(m-1)](k+1)^2, \quad L = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad L/q \geq \delta k/2. \quad (\text{CJ.11})$$

The original Hermitian compression proof gives $0 < L \leq \epsilon_N$ at every admitted degree: on the full sum of generalized eigenspaces with $\Re \lambda > k/2$, its trace is L . An orthogonal projection onto that subspace in the original metric has trace pairing at most ϵ_N with a Hermitian matrix of eigenvalues $\epsilon_N, -\epsilon_N$ and zeros. This is the same source metric as (CJ.2). Thus each contraction occurring in (CJ.9)–(CJ.10) is strictly between zero and one. A value one for any δ_N would make a positive radius product zero.

For either original length- q block $(i, j) = (q-1, 2q-1)$ or $(q, 2q)$, taking logarithms of its exact product now gives

$$\log(V_i/V_j) = \sum_{N=i}^{j-1} \log(\epsilon_N^2 + \phi_N^2) - \log(\omega_j/\omega_i) + \mathcal{P}_{i,j}, \quad (\text{CJ.12})$$

where the entire contraction penalty is

$$\begin{aligned} \mathcal{P}_{q-1, 2q-1} &= -\log(1 - \delta_{2q-1}) - 2 \sum_{N=q}^{2q-2} \log(1 - \delta_N), \\ \mathcal{P}_{q, 2q} &= -\log(1 - \delta_q) - \log(1 - \delta_{2q}) - 2 \sum_{N=q+1}^{2q-1} \log(1 - \delta_N). \end{aligned} \quad (\text{CJ.13})$$

Both sums retain their literal first and last degrees and are nonnegative. Since $\epsilon_N \geq L$, the exact finite bound is

$$\log(V_i/V_j) \geq 2q \log L - \log(\omega_j/\omega_i) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L^2). \quad (\text{CJ.14})$$

Every phase term remains; none is assigned the value zero.

3 Actual comparable norm windows with complete constants

The original arithmetic lower-envelope proof gives, for $k \geq 3$,

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h. \quad (\text{CJ.15})$$

Its complete analytic proof is retained in the arithmetic source appendix and AU.1–AU.12. Fix $0 < b < \pi/2$, and keep $M_h(\pm b) = \int e^{\pm bt} w_h(t) dt$. Set the explicit positive constants

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \frac{1}{2} e^{-\alpha} \sqrt{c_0 t_h} 2^{-B/2}, \\ M &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{M}/b. \end{aligned} \quad (\text{CJ.16})$$

For every integer $n \geq k \geq 3$ the literal monic norms obey

$$(\ell_h n)^{2n} \leq \omega_n \leq (U_h n)^{2n}. \quad (\text{CJ.17})$$

Here is the full envelope calculation. The exact minimum squared norm of a monic polynomial on $[-n, n]$ is

$$\frac{2n^{2n+1}}{2n+1} \left[\frac{2^n (n!)^2}{(2n)!} \right]^2 \geq \frac{2}{3} n^{2n} 4^{-n}.$$

The last inequality uses $(2n)!/(n!)^2 \leq 4^n$ and $2n/(2n+1) \geq 2/3$. The original chart's factor i^n has modulus one; its monic polynomial space is exactly this extremal space through the inverse substitution. On $[-n, n]$, (CJ.15) is at least $c_h \vartheta_h^{k-3} e^{-\alpha(k-3+n)} (k-2+n)^{-B}$. Use, in their stated directions,

$$\vartheta_h^{k-3} \geq t_h^n, \quad e^{-\alpha(k-3+n)} \geq e^{-2\alpha n}, \quad (k-2+n)^{-B} \geq (2n)^{-B} \geq 2^{-nB}, \quad 2c_h/3 \geq c_0^n.$$

The elementary $2n \leq 2^n$ holds at $n = 1$ and follows by induction from $2(n+1) \leq 4n$ for $n \geq 1$. Multiplication proves the lower bound (CJ.17). For the upper bound use the monic trial polynomial $(S - c)^n$, the nonnegative exponential series, and convolution to obtain $\omega_n \leq (2n)! b^{-2n} (M_h(b)^k + M_h(-b)^k)$. This is at most $(2n)^{2n} b^{-2n} 2M^n$, which is at most $(4\sqrt{M}n/b)^{2n}$ since $2 \leq 4^n$. This proves both inequalities with the original mass retained.

For integers $n \geq k \geq 3$ and $n/2 \leq r \leq 2n$, divide the two instances of (CJ.17) and take the positive $2r$ -th root:

$$\begin{aligned} \left(\frac{\omega_{n+r}}{\omega_n} \right)^{1/(2r)} &\leq n U_h^{1+n/r} \ell_h^{-n/r} (1 + r/n)^{1+n/r} \leq C_h^{\text{win}} n, \\ C_h^{\text{win}} &= 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2. \end{aligned} \quad (\text{CJ.18})$$

Indeed $1 + n/r \leq 3$, $n/r \leq 2$, and $1 + r/n \leq 3$. For each block above, $r = q$ and $n = q$ or $q - 1$. Equation (CJ.11) gives $q - 1 \geq k$ for $k \geq 3$, and $q/2 \leq q \leq 2(q - 1)$ for these q . Both norm factors are therefore at most $C_h^{\text{win}} q$. Combining (CJ.11), (CJ.14) and (CJ.18), with $D_h = \delta/(2C_h^{\text{win}}) > 0$, proves for each block

$$\log(V_i/V_j) \geq 2q \log(D_h k) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L^2). \quad (\text{CJ.19})$$

Adding the two bounds and dividing by $q \log k$ proves the four-volume lower limit at least four. This includes the complete finite penalties before taking that consequence.

4 The full relation cost and its retained zero directions

For any admitted $i < j$, set $F = [b_{i+1}, \dots, b_j]$ and $\Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j)$. The exact update is $K_j - K_i = F\Omega^{-1}F^*$, so

$$V_i/V_j = \det(I + F\Omega^{-1}F^*G_i) = \det(I + \Omega^{-1}F^*G_iF). \quad (\text{CJ.20})$$

The rectangular determinant equality follows by eliminating either off-diagonal block of the original block matrix

$$\begin{pmatrix} I & F \\ -\Omega^{-1}F^*G_i & I \end{pmatrix}.$$

On the new-column source $\mathbb{C}^{j-i}$ with metric Ω , the second cost is positive self-adjoint. Its eigenvalues are nonnegative, including their zero multiplicities. Write $L_{i,j} : \mathbb{C}^{i+1} \rightarrow \mathbb{C}^{j+1}$ for padding in the same monic orthogonal basis, and $T_{\text{new}} : \mathbb{C}^{j-i} \rightarrow \mathbb{C}^{j+1}$ for the coordinate inclusion into positions $i+1, \dots, j$. The original relation map from $(\mathbb{C}^{j-i}, \Omega)$ to $(\mathcal{P}_j, O_j)$ is $\mathfrak{b} = T_{\text{new}} - L_{i,j}R_iF$. The new columns and the lower source are orthogonal, and $R_i^*O_iR_i = G_i$, so

$$B_j\mathfrak{b} = 0, \quad \mathfrak{b}^*O_j\mathfrak{b} = \Omega + F^*G_iF. \quad (\text{CJ.21})$$

Set $\mathcal{D}_N = \ker B_N = \chi\mathcal{P}_{N-q}$, with the zero space when $N = q-1$. The new-coordinate component of $\mathfrak{b}$ is the identity; it is therefore injective. Its image lies in $\mathcal{D}_j \cap (LD_i)^\perp$, whose dimension is $j-i$, by the dimensions of the two nested relation spaces. Thus it is an isomorphism onto that entire relation layer. Reduction sends its labelled vectors to their receiving supported zero; the lifted reduction fixes external absence τ .

At zero tilt the full quartet gives an even original density. Its u -polynomial $\psi = i^{-q}\chi(c + iu)$ satisfies $\psi(-u) = (-1)^q\psi(u)$ because its complete root multiset is stable under negation. Uniqueness of the monic orthogonal polynomial gives $Q_n(-u) = (-1)^nQ_n(u)$: substituting $-u$ preserves all original integral pairings and the adjusted leading coefficient. Monic division modulo ψ preserves parity. When k is even, q in (CJ.11) is odd. In the first block the q new polynomials $Q_q, \dots, Q_{2q-1}$ contain $(q+1)/2$ odd polynomials, while the odd quotient subspace has dimension $(q-1)/2$. Their reduction columns are dependent. The original $S = c + iu$ basis map and the nonzero column phases i^n preserve rank. Thus at least one of the q cost eigenvalues in that block is zero. It remains in the full determinant as a factor one; (CJ.19)–(CJ.21) remain valid with every original dimension and nilpotent direction retained.

For either length- q block let $\lambda_a \geq 0$ be all its cost eigenvalues. Equations (CJ.19)–(CJ.20) imply

$$\prod_{a=1}^q (1 + \lambda_a) \geq (D_h k)^{2q}, \quad \sum_{a=1}^q (1 + \lambda_a)^s \geq q(D_h k)^{2s} \quad (s > 0). \quad (\text{CJ.22})$$

The second assertion is the arithmetic-geometric mean inequality for the q positive numbers $(1 + \lambda_a)^s$; it also obeys the trivial lower bound q . Here is its exact restriction map. Give the same coefficient space E its two original metrics and let $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$ be the identity coefficient map. Its Hilbert adjoint is $T_{i,j} : (E, G_j) \rightarrow (E, G_i)$ with matrix $G_i^{-1}G_j$, because $x^*G_jy = x^*G_i(G_i^{-1}G_j)y$ for every x, y . Thus $Q = T_{i,j}I_{i,j}$ acts on (E, G_i) and satisfies

$$Q^{-1} = G_j^{-1}G_i = K_jG_i = I + F\Omega^{-1}F^*G_i.$$

It is positive self-adjoint in that original metric; its quadratic form is $x^*G_iK_jG_ix > 0$ for $x \neq 0$. Consequently its eigenvalues are exactly the $1 + \lambda_a$ in (CJ.22), with all zero cost directions giving eigenvalue one. Hence (CJ.22) is also an inverse-restriction trace bound in its specified original metric. No individual positive cost lower bound or strict-positive rank claim is inferred.

5 Provenance and verification scope

The complete cumulative sources CV and TVB prove the radius and phase map used here, and the complete arithmetic source and AU prove the actual density envelope (CJ.15). PR 25 supplies the comparable-window constants and first-degree propagation. The full first-window phase product (CJ.9), its finite penalties (CJ.13)–(CJ.19), and their connection to the retained relation costs are written out here. The owner review at the cited head requires a nonnegative-eigenvalue correction and workflow-trigger corrections; this chapter preserves that status and the complete proposed parity proof. It performs no local Lean execution and makes no new formal-verification claim. Its finite algebra and analytic envelope calculations belong to this written proof.